

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

The goal of this program is to observe transiting exoplanets by measuring the light curve created by the planet passing in front of the star using relative photometry. Repeating measurements of this target allows us to further confirm that the transiting body is indeed an exoplanet by ruling out the possibility that the curve is caused by a star passing behind the primary star as well as providing another set of data that further research can use to refine the error on the measurements via statistical analysis. To do so requires detecting changes in the apparent brightness of the star up to a tenth of a percent, implying a large signal-to-noise ratio. Our primary target is the transit of TOI-1516 b, with a backup target of HD 209458 b.

Telescope Requested (0.5 points): Kuiper 61" Telescope

Instrument (0.5 points): Mont4k CCD Imager

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

The RA of the moon compared to that of the stars is far enough away and our targets are bright enough that the moon should have negligible impact. Furthermore, because we are comparing the brightness of the star to that of the stars around it, any additional light from the moon will be approximately the same for both target and background, leaving the relative difference unchanged. The moon will be at approximately 52% of its maximum brightness for our first target and approximately 41% of its maximum brightness for the second.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
TOI-1516 b	340.08462 degrees	69.50373 degrees	10.9	exoplanet	October 2, 2026
HD 209458 b	330.79504 degrees	18.88424 degrees	7.7	exoplanet	October 3, 2026

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Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

Due to the relatively similar sizes of the smallest possible stars and largest possible planets, we need a signal-to-noise ratio of at least 1000. For target TOI-1516 b, we require approximately three hours and forty-five minutes from around 11:15pm to around 3:00am the night of October 2nd. For target HD 209458 b, we require approximately four hours from around 10:00pm to around 2:00am the night of October 3rd. Our exposure time per snapshot is thirty seconds, for a signal-to-noise ratio of roughly 1046.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

These observations will repeat previous measurements of these targets. Recreating previous results is integral to the scientific method, and doing so here will further confirm that these transits result from an exoplanet by adding another set of data, with which statistical analysis can be used to refine the error on previous results.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If neither our primary nor our secondary target can be observed, we can use archival data of the systems instead to examine their light curves and understand how they came to be understood as exoplanet transits.